

# Laboratorio07

## Resoluções - Lab 07

01) Baixe o arquivo `uwmadison.sqlite3`. Conecte-se a ele usando o pacote `RSQLite`, armazenando a conexão em uma variável `conn`

```
library(RSQLite)

conn <- dbConnect(SQLite(), "database.sqlite3")

# conferindo as tabelas disponíveis
dbListTables(conn)
```

```
[1] "course_offerings"    "courses"             "grade_distributions"
[4] "instructors"         "rooms"                "schedules"
[7] "sections"           "subject_memberships" "subjects"
[10] "teachings"
```

02) Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (`subjects.abbreviation='STAT'`)

```
query_profs <- "
SELECT DISTINCT i.id, i.name
FROM instructors i
INNER JOIN teachings t      ON t.instructor_id = i.id
INNER JOIN sections s      ON s.uuid = t.section_uuid
INNER JOIN course_offerings co ON co.uuid = s.course_offering_uuid
INNER JOIN subject_memberships sm ON sm.course_offering_uuid = co.uuid
INNER JOIN subjects sub    ON sub.code = sm.subject_code
WHERE sub.abbreviation = 'STAT'
"

profs_stat <- dbGetQuery(conn, query_profs)

nrow(profs_stat) # quantos são
```

```
[1] 379
```

```
profs_stat # quem são
```

|   | id      | name            |
|---|---------|-----------------|
| 1 | 4392906 | MINJING TAO     |
| 2 | 393380  | DONALD PORTER   |
| 3 | 4146030 | SHENG WANG      |
| 4 | 5697552 | KUNLING HUANG   |
| 5 | 4898014 | DONG XIA        |
| 6 | 4613728 | JENNIFER NGUYEN |
| 7 | 5520811 | SEULKEE YUN     |
| 8 | 5516726 | DUZHE WANG      |

|    |         |                      |
|----|---------|----------------------|
| 9  | 5696436 | YONGFENG WU          |
| 10 | 4636315 | JARED BROWN          |
| 11 | 5700058 | YUCHEN ZHOU          |
| 12 | 4641200 | VICTOR LUO           |
| 13 | 4884161 | GUN WOONG PARK       |
| 14 | 5011312 | BROOK LUERS          |
| 15 | 4728844 | DUY NGUYEN           |
| 16 | 608997  | JOHN GILLET          |
| 17 | 4161984 | QUOC TRAN            |
| 18 | 4576059 | YAOLAO XU            |
| 19 | 4436368 | JIE ZHANG            |
| 20 | 5015113 | LONG PHAN            |
| 21 | 5161294 | WESLEY CHANG         |
| 22 | 5329574 | TONG LI              |
| 23 | 982957  | SHANE HUBLER         |
| 24 | 4437355 | FAN YANG             |
| 25 | 4731731 | JU HEE CHO           |
| 26 | 4890283 | JEEA CHOI            |
| 27 | 4885477 | SHIXUE LIU           |
| 28 | 2602108 | WEI-YIN LOH          |
| 29 | 4155990 | BIN DAI              |
| 30 | 4145889 | XIPEI YANG           |
| 31 | 2602050 | DOUGLAS M. BATES     |
| 32 | 783769  | MICHAEL GEORGE ILLIS |
| 33 | 4845798 | LIAM JOHNSTON        |
| 34 | 5498648 | CRYSTAL CHEN         |
| 35 | 5693277 | YIFAN MEI            |
| 36 | 5690182 | RUI CHEN             |
| 37 | 4734274 | JILI WANG            |
| 38 | 5017774 | LUWAN ZHANG          |
| 39 | 5017655 | GINA BENNINGER       |
| 40 | 5410648 | JIE SONG             |
| 41 | 3791802 | XIWEN MA             |
| 42 | 3456935 | HAODA FU             |
| 43 | 3773167 | DEYUAN JIANG         |
| 44 | 3973318 | JINGJIANG PENG       |
| 45 | 3566378 | ERICA LEE DEADMAN    |
| 46 | 3775809 | AKICHIKA OZEKI       |
| 47 | 4280220 | MENG SONG            |
| 48 | 5520701 | HAO TENG             |
| 49 | 4897725 | DONGGYU KIM          |
| 50 | 5332940 | COLIN LONGHURST      |
| 51 | 361804  | FREDERICK BOEHM      |
| 52 | 5518378 | YONGSU LEE           |
| 53 | 5320625 | GONZALO CONTADOR     |
| 54 | 5514514 | HAO CHEN             |
| 55 | 4588103 | XIN ZHANG            |
| 56 | 5632570 | BO YANG              |
| 57 | 5898424 | RUNGANG HAN          |
| 58 | 5068296 | SEAN KENT            |
| 59 | 6043299 | DEBRAJ DAS           |
| 60 | 5327139 | YI LI                |
| 61 | 5887049 | CLAIRE BOBST         |
| 62 | 5883206 | CHAN PARK            |

|     |         |                         |
|-----|---------|-------------------------|
| 63  | 5323628 | SIJING LI               |
| 64  | 5017571 | SHENGJI JIA             |
| 65  | 5903922 | YUCHANG WU              |
| 66  | 3626007 | ZHUANG WU               |
| 67  | 5170801 | XIAOMAO LI              |
| 68  | 5321537 | HYEBIN SONG             |
| 69  | 5326059 | YOURAN QI               |
| 70  | 5177493 | YOUNG MIN PARK          |
| 71  | 5186047 | BIN ZHANG               |
| 72  | 4582107 | QIURONG CUI             |
| 73  | 4733192 | SEHO PARK               |
| 74  | 4285558 | SHENG ZHANG             |
| 75  | 5168069 | HUIKUN ZHANG            |
| 76  | 5162339 | ALYSSA DIGILIO          |
| 77  | 5214858 | YUQING XU               |
| 78  | 4538540 | STEPHEN BERG            |
| 79  | 5326263 | PEIGEN ZHOU             |
| 80  | 4721620 | YAN CHEN                |
| 81  | 4460098 | QIONG ZHANG             |
| 82  | 4278664 | YOUNGDEOK HWANG         |
| 83  | 5010171 | HAN CHEN                |
| 84  | 640407  | NICHOLAS STEPHEN KEULER |
| 85  | 3107260 | ZHENGXIAO WU            |
| 86  | 4144479 | YUJIN CHUNG             |
| 87  | 4590481 | WENWEN ZHANG            |
| 88  | 4149787 | PERLA REYES             |
| 89  | 3974100 | QI TANG                 |
| 90  | 3629214 | CHIA-CHIEH LIN          |
| 91  | 4433317 | JIALE XU                |
| 92  | 4285354 | YANG ZHAO               |
| 93  | 3620612 | XINWEI DENG             |
| 94  | 4133622 | MING XIE                |
| 95  | 3783649 | JINGCI MENG             |
| 96  | 143932  | ISMOR FISCHER           |
| 97  | 4590715 | JIAJIE CHEN             |
| 98  | 4138542 | ZUOFENG SHANG           |
| 99  | 4435790 | HAO ZHENG               |
| 100 | 4590559 | XU XU                   |
| 101 | 4436066 | BIN ZHU                 |
| 102 | 4432994 | QUEFENG LI              |
| 103 | 3961448 | YU QIU LIU              |
| 104 | 3981051 | JUN LI                  |
| 105 | 5204089 | FAN GAO                 |
| 106 | 4788841 | VICTORIA MANSFIELD      |
| 107 | 5498092 | ANQI SHI                |
| 108 | 5317373 | KYLE HEBERT             |
| 109 | 5322566 | XIUYU MA                |
| 110 | 4597046 | CHENLIANG XU            |
| 111 | 3976099 | CHANHAN HSU             |
| 112 | 4906458 | YUANZHI LI              |
| 113 | 2602032 | KAM-WAH TSUI            |
| 114 | 4430246 | XINYU SONG              |
| 115 | 4879129 | TIMOTHY IDOWU           |
| 116 | 4905505 | XIAO NIE                |

|     |         |                       |
|-----|---------|-----------------------|
| 117 | 4586558 | CHEN CHENG            |
| 118 | 5003565 | ANDREW LESLIE         |
| 119 | 4895244 | YAOGUO XIE            |
| 120 | 4744350 | KEVIN PACKARD         |
| 121 | 4586128 | YI LIU                |
| 122 | 5322094 | WENZHI CAO            |
| 123 | 4610245 | JOSEPH DEUTSCH        |
| 124 | 5521838 | LAN LUO               |
| 125 | 3727371 | RYAN ZEA              |
| 126 | 2601091 | ROBERT WARDROP        |
| 127 | 3981781 | SANGBUM CHOI          |
| 128 | 4285528 | XIN LI                |
| 129 | 3573852 | SEUNGBONG HAN         |
| 130 | 3339239 | ZHIGUO XIAO           |
| 131 | 2601613 | RICHARD A. JOHNSON    |
| 132 | 4128750 | RUI TANG              |
| 133 | 801445  | IAN BRANSTAD RILEY    |
| 134 | 3484236 | BO HUANG              |
| 135 | 4149843 | HEATHER MARIE BRAZEAU |
| 136 | 4286138 | XU HE                 |
| 137 | 495087  | GARY HOWARD SCHROEDER |
| 138 | 3781764 | JUN ZHANG             |
| 139 | 3973813 | CHENXI LI             |
| 140 | 3955531 | SONA Z SWANSON        |
| 141 | 3791787 | YUAN JIANG            |
| 142 | 3620552 | JUNHEE HAN            |
| 143 | 4578330 | LILUN DU              |
| 144 | 5170924 | XIRAN WANG            |
| 145 | 5099221 | ZIFENG ZHAO           |
| 146 | 5176022 | CUIZE HAN             |
| 147 | 5170492 | YING ZHANG            |
| 148 | 4880968 | JOHN DAVIS            |
| 149 | 5018463 | CHEN JING             |
| 150 | 4902482 | JINGLAN LI            |
| 151 | 4285538 | YI CHAI               |
| 152 | 4904255 | THU LE                |
| 153 | 4958796 | ALAN HUANG            |
| 154 | 4725072 | HAOYANG FAN           |
| 155 | 5690247 | BOWEN HU              |
| 156 | 5664494 | TZU HSIANG HUNG       |
| 157 | 5567363 | TUN LEE NG            |
| 158 | 4787911 | ABIGAIL BENZINE       |
| 159 | 5656034 | ALEXANDER COVINGTON   |
| 160 | 3376532 | CHELSEY GREEN         |
| 161 | 5708167 | MICHAEL KUTZLER       |
| 162 | 5693114 | LILI ZHENG            |
| 163 | 5488796 | REBECCA POST          |
| 164 | 5010560 | LU YANG               |
| 165 | 5020061 | RUOSI GUO             |
| 166 | 3668575 | XUN ZHAO              |
| 167 | 3965653 | MIN NIU               |
| 168 | 4603165 | JIAN WU               |
| 169 | 3780817 | WEI ZHENG             |
| 170 | 4726666 | XINJIE HE             |

|     |         |                              |
|-----|---------|------------------------------|
| 171 | 4292460 | JIE WEI                      |
| 172 | 4721578 | CLAUDIA SOLIS LEMUS          |
| 173 | 4726703 | XUEYAO CHEN                  |
| 174 | 5302154 | MANJUSHA KANCHARLA           |
| 175 | 4947463 | MITCHELL PAUKNER             |
| 176 | 5258581 | XIAOWU DAI                   |
| 177 | 5906594 | SHENG WANG                   |
| 178 | 5592111 | CHENGNING ZHANG              |
| 179 | 5862942 | GINA OH                      |
| 180 | 5192960 | LUXI CAO                     |
| 181 | 4953246 | SONG WANG                    |
| 182 | 4435446 | LAM HO                       |
| 183 | 4896939 | TIEN VO                      |
| 184 | 4598772 | SHIZHEN WANG                 |
| 185 | 5176905 | SOHEIL SADEGHI               |
| 186 | 5144884 | BINGYING XIE                 |
| 187 | 4957688 | GENG LI                      |
| 188 | 4636145 | JENNIFER BIRSTLER            |
| 189 | 4195000 | MARIA KAMENETSKY             |
| 190 | 5478075 | KYLE HERBET                  |
| 191 | 4886987 | JIANCHANG HU                 |
| 192 | 4896172 | TRAM TA                      |
| 193 | 4733966 | YUAN WANG                    |
| 194 | 3777992 | KAZUHIKO SHINKI              |
| 195 | 3622809 | SIJIAN WANG                  |
| 196 | 3613299 | CHIEN-WEI CHEN               |
| 197 | 5852265 | BEHZAD AALIPUR               |
| 198 | 4264265 | DEREK BEAN                   |
| 199 | 4140469 | BEN ADAM HAALAND             |
| 200 | 2824494 | ELOISA D CHAVAS              |
| 201 | 3800708 | YALI WANG                    |
| 202 | 3959251 | VARSHA KULKARNI              |
| 203 | 3253489 | SCOTT JOSEPH HETZEL          |
| 204 | 3575433 | MIN JUNG LEE                 |
| 205 | 3460189 | FANG FANG                    |
| 206 | 4199106 | KRISTEN CYFFKA               |
| 207 | 3351080 | TING-LI LIN                  |
| 208 | 751075  | BRET LARGET                  |
| 209 | 4491860 | KATHERINE GOODE              |
| 210 | 2602042 | ERIK NORDHEIM                |
| 211 | 5523214 | NING FAN                     |
| 212 | 4646729 | ANRU ZHANG                   |
| 213 | 5214164 | GARVESH RASKUTTI             |
| 214 | 5476019 | KRISHNAKUMAR BALASUBRAMANIAN |
| 215 | 4695492 | YAZHEN WANG                  |
| 216 | 4712098 | CHAOYANG YU                  |
| 217 | 5323691 | YILIN ZHANG                  |
| 218 | 4108525 | ZHENGJUN ZHANG               |
| 219 | 4194670 | XIUFENG SHAO                 |
| 220 | 5167973 | YUAN LI                      |
| 221 | 4753018 | BRET HANLON                  |
| 222 | 3980675 | LILI LAN                     |
| 223 | 3955546 | CECILE ANE                   |
| 224 | 1426304 | ZHANG CHUNMING               |

|     |         |                               |
|-----|---------|-------------------------------|
| 225 | 3443667 | LEI XU                        |
| 226 | 5514561 | MUHONG GAO                    |
| 227 | 4963378 | AUGUST JENSEN                 |
| 228 | 4591139 | XIAO GUO                      |
| 229 | 2601039 | THOMAS G. KURTZ               |
| 230 | 3523865 | NICHOLAS HENDERSON            |
| 231 | 4903489 | CHENSHENG KUANG               |
| 232 | 4590738 | TAERI UHM                     |
| 233 | 3882760 | NORBERT BINKIEWICZ            |
| 234 | 4929500 | KARL ROHE                     |
| 235 | 3095399 | JUNGWON MUN                   |
| 236 | 1126199 | JOSEP GINEBRA                 |
| 237 | 3781422 | TAO YU                        |
| 238 | 4286679 | XINXIN YU                     |
| 239 | 5323690 | HAO ZHOU                      |
| 240 | 4277002 | LIE XIONG                     |
| 241 | 3338237 | QIAN ZHIGUANG                 |
| 242 | 5515201 | SHANG WU                      |
| 243 | 5905014 | SHAN LU                       |
| 244 | 3122658 | STEPHEN AARON STANHOPE        |
| 245 | 3103459 | LANCINE KONATE                |
| 246 | 4886028 | GUILHERME VIEIRA NUNES LUDWIG |
| 247 | 5631398 | PAUL SAVARIAPPAN              |
| 248 | 5015194 | CUICUI QI                     |
| 249 | 3404299 | KJELL DOKSUM                  |
| 250 | 5305675 | DEREK NORTON                  |
| 251 | 4261114 | KEEGAN KORTHAUER              |
| 252 | 5012817 | KAILEI CHEN                   |
| 253 | 5023414 | THEVAASIINEN CHANDERENG       |
| 254 | 3806328 | JOHN KANE                     |
| 255 | 4593965 | SOKOL VAKO                    |
| 256 | 5718599 | QING LI                       |
| 257 | 5514517 | YOUNG LEE                     |
| 258 | 4734607 | HYUNSEUNG KANG                |
| 259 | 5694563 | BEHZAD AALIPUR HAFSHEJANI     |
| 260 | 5170885 | SHULEI WANG                   |
| 261 | 5693629 | ZHONGJIE YU                   |
| 262 | 5514932 | LIN QI                        |
| 263 | 4286103 | JIWEI ZHAO                    |
| 264 | 3801422 | SUNDUZ KELES                  |
| 265 | 4588354 | SHUYUN YE                     |
| 266 | 3483534 | HUI WANG                      |
| 267 | 3978606 | KEVIN HASEGAWA ENG            |
| 268 | 3343005 | XIAODAN WEI                   |
| 269 | 5167906 | MARIA ROJO                    |
| 270 | 3343055 | RUIFENG XU                    |
| 271 | 5644228 | YONGJOON KIM                  |
| 272 | 232828  | NELLIE LAUGHLIN               |
| 273 | 270258  | MICHAEL HOGAN                 |
| 274 | 4957916 | TIMOTHY IDOWU                 |
| 275 | 4541610 | KARL BROMAN                   |
| 276 | 789363  | REBECCA KOSCIK                |
| 277 | 5887858 | RONGJUN ZHU                   |
| 278 | 5906523 | BI CHENG WU                   |

|     |         |                        |
|-----|---------|------------------------|
| 279 | 5907599 | YUNONG LIN             |
| 280 | 4588525 | ZHIGENG GENG           |
| 281 | 4729481 | XIAOPING FENG          |
| 282 | 4271924 | SHUANG HUANG           |
| 283 | 4436002 | GUANNAN SUN            |
| 284 | 4140397 | JEE YEON KIM           |
| 285 | 5127444 | GREGORY SHINAULT       |
| 286 | 3230527 | TONGHAI YANG           |
| 287 | 3374880 | TIMO SEPPALAINEN       |
| 288 | 4538519 | BENEDEK VALKO          |
| 289 | 2602109 | DAVID GRIFFEATH        |
| 290 | 4308515 | DAVID ANDERSON         |
| 291 | 1515839 | PHILIP WOOD            |
| 292 | 4601948 | FLORIAN BERTRAND       |
| 293 | 3956063 | JASON RICHARD SWANSON  |
| 294 | 3573775 | ALEXANDER KISELEV      |
| 295 | 4752238 | SAMUEL STECHMANN       |
| 296 | 4434130 | WAI TONG FAN           |
| 297 | 3620114 | JUN YIN                |
| 298 | 2600973 | STEFFEN LEMPP          |
| 299 | 3987897 | RUIFANG SONG           |
| 300 | 2601633 | JAMES D KUELBS         |
| 301 | 129415  | ANATOLE BECK           |
| 302 | 5620719 | DONGHYUN LEE           |
| 303 | 3674168 | ANDREJ ZLATOS          |
| 304 | 5846795 | DANIELE CAPPELLETTI    |
| 305 | 6029366 | MIHAELA IFRIM          |
| 306 | 5110390 | SEBASTIEN ROCH         |
| 307 | 5354849 | SCOTT HOTTOVY          |
| 308 | 1634418 | JONATHON PETERSON      |
| 309 | 2601051 | STEPHEN WAINGER        |
| 310 | 4680082 | ALEXANDER FISH         |
| 311 | 4761200 | GREGORIO MORENO-FLORES |
| 312 | 3117835 | SUKHENDU MEHROTRA      |
| 313 | 2600233 | DIETRICH UHLENBROCK    |
| 314 | 3437432 | LEV BORISOV            |
| 315 | 2600141 | RICHARD A BRUALDI      |
| 316 | 2600893 | ARNOLD MILLER          |
| 317 | 3467530 | MATTHEW BALLARD        |
| 318 | 3661782 | ALBRECHT KLEMM         |
| 319 | 2601093 | PAUL M TERWILLIGER     |
| 320 | 5452538 | STEVEN SAM             |
| 321 | 3127329 | KEN ONO                |
| 322 | 4957895 | ROBERT HARRON          |
| 323 | 4172260 | YANNAN QIU             |
| 324 | 4896511 | JOHN WILTSHIRE-GORDON  |
| 325 | 207204  | KYUNGMAH KIM           |
| 326 | 3855025 | CHRISTOPHER WAGNER     |
| 327 | 967335  | PAUL RATHOUZ           |
| 328 | 3382451 | STEPHEN WRIGHT         |
| 329 | 4632211 | BENJAMIN RECHT         |
| 330 | 1114818 | MICHAEL FERRIS         |
| 331 | 3627934 | JESSE THOMAS HOLZER    |
| 332 | 5167990 | HUILIN HU              |

|     |         |                              |
|-----|---------|------------------------------|
| 333 | 5451706 | ALBERTO DEL PIA              |
| 334 | 2601512 | ROBERT R MEYER               |
| 335 | 3127330 | SHI JIN                      |
| 336 | 333334  | MARY LINDSTROM               |
| 337 | 3111842 | ADIN-CRISTIAN ANDREI         |
| 338 | 5840930 | MICHAEL LIOU                 |
| 339 | 4724058 | CHAOQUN MEI                  |
| 340 | 340544  | CHRISTINA M. KENDZIORSKI     |
| 341 | 893827  | RONALD GANGNON               |
| 342 | 3780010 | ROBERT WAYNE GREEN           |
| 343 | 4312759 | COLE COOK                    |
| 344 | 2601621 | MURRAY CLAYTON               |
| 345 | 4633355 | QI JIANG                     |
| 346 | 321374  | JUN ZHU                      |
| 347 | 4952769 | ANNE BRUCKNER                |
| 348 | 5520995 | TING YE                      |
| 349 | 1079018 | JUN SHAO                     |
| 350 | 3623432 | HUAIBAO FENG                 |
| 351 | 5591917 | ZIJIAN NI                    |
| 352 | 417071  | THOMAS COOK                  |
| 353 | 2601926 | RICHARD J. CHAPPELL          |
| 354 | 3404298 | MOO K CHUNG                  |
| 355 | 2601916 | MICHAEL NEWTON               |
| 356 | 2602055 | BRIAN YANDELL                |
| 357 | 394810  | MICHELLE HARRIS              |
| 358 | 2600689 | NORMAN DRAPER                |
| 359 | 3010828 | YU MENGANG                   |
| 360 | 3077702 | MING YUAN                    |
| 361 | 3783301 | FANGFANG WANG                |
| 362 | 4541633 | JUNHO LEE                    |
| 363 | 2600488 | GRACE WAHBA                  |
| 364 | 3011936 | JASON P FINE                 |
| 365 | 989414  | MICHAEL RENE KOSOROK         |
| 366 | 4426932 | YAJUAN SI                    |
| 367 | 4726678 | LU MAO                       |
| 368 | 2601596 | DAVID DEMETS                 |
| 369 | 5515948 | JAMES ANDERSON               |
| 370 | 5189535 | EDWARD ERKER                 |
| 371 | 4699427 | SHUAI CHEN                   |
| 372 | 2601674 | MARI PALTA                   |
| 373 | 4272783 | GUANHUA CHEN                 |
| 374 | 3884290 | SRIKANTHMADHAVAN ARAVAMUTHAN |
| 375 | 1505727 | NATALIA DE LEON GATTI        |
| 376 | 1505756 | GUILHERME ROSA               |
| 377 | 3457484 | YANBING ZHENG                |
| 378 | 138086  | CHRISTINE SORKNESS           |
| 379 | 245914  | MARIAN R FISHER              |

03)

a.

```
query_prof_gpa <- "
SELECT
```

```

i.id AS instructor_id,
i.name AS instructor_name,
gd.a_count, gd.ab_count, gd.b_count, gd.bc_count,
gd.c_count, gd.d_count, gd.f_count
FROM grade_distributions gd
INNER JOIN course_offerings co ON co.uuid = gd.course_offering_uuid
INNER JOIN subject_memberships sm ON sm.course_offering_uuid = co.uuid
INNER JOIN subjects sub ON sub.code = sm.subject_code
INNER JOIN sections s ON s.course_offering_uuid = co.uuid
                        AND s.number = gd.section_number
INNER JOIN teachings t ON t.section_uuid = s.uuid
INNER JOIN instructors i ON i.id = t.instructor_id
WHERE sub.abbreviation = 'STAT'
"

dados_prof <- dbGetQuery(conn, query_prof_gpa)

```

b.

```

query_disc_gpa <- "
SELECT
  c.uuid AS course_uuid,
  c.number AS course_number,
  c.name AS course_name,
  gd.a_count, gd.ab_count, gd.b_count, gd.bc_count,
  gd.c_count, gd.d_count, gd.f_count
FROM grade_distributions gd
INNER JOIN course_offerings co ON co.uuid = gd.course_offering_uuid
INNER JOIN courses c ON c.uuid = co.course_uuid
INNER JOIN subject_memberships sm ON sm.course_offering_uuid = co.uuid
INNER JOIN subjects sub ON sub.code = sm.subject_code
WHERE sub.abbreviation = 'STAT'
"

dados_disc <- dbGetQuery(conn, query_disc_gpa)

```

c.

Observação: Nessa questão precisamos conferir como dados\_prof estava chegando, como estava como numerico foi necessario fazer uma conversao.

```
str(dados_prof)
```

```

'data.frame': 1820 obs. of 9 variables:
 $ instructor_id : chr "393380" "393380" "4898014" "4636315" ...
 $ instructor_name: chr "DONALD PORTER" "DONALD PORTER" "DONG XIA" "JARED BROWN" ...
 $ a_count       : chr "17" "21" "15" "11" ...
 $ ab_count      : chr "14" "16" "11" "15" ...
 $ b_count       : chr "27" "23" "27" "23" ...
 $ bc_count      : chr "7" "6" "7" "20" ...
 $ c_count       : chr "20" "13" "11" "8" ...
 $ d_count       : chr "2" "1" "3" "7" ...
 $ f_count       : chr "0" "2" "0" "0" ...

```

```
cols_notas <- c("a_count", "ab_count", "b_count", "bc_count",
               "c_count", "d_count", "f_count")

dados_prof[cols_notas] <- lapply(dados_prof[cols_notas], as.numeric)

str(dados_prof) # conferir que virou "num" agora
```

```
'data.frame': 1820 obs. of 9 variables:
 $ instructor_id : chr  "393380" "393380" "4898014" "4636315" ...
 $ instructor_name: chr  "DONALD PORTER" "DONALD PORTER" "DONG XIA" "JARED BROWN" ...
 $ a_count       : num  17 21 15 11 15 27 22 25 45 51 ...
 $ ab_count      : num  14 16 11 15 3 5 11 11 11 16 ...
 $ b_count       : num  27 23 27 23 28 13 11 19 19 12 ...
 $ bc_count      : num  7 6 7 20 6 14 13 3 3 7 ...
 $ c_count       : num  20 13 11 8 11 17 14 7 6 5 ...
 $ d_count       : num  2 1 3 7 1 3 3 0 1 1 ...
 $ f_count       : num  0 2 0 0 2 2 5 1 2 0 ...
```

```
str(dados_disc)
```

```
'data.frame': 1716 obs. of 10 variables:
 $ course_uuid : chr  "b013a134-06a6-31a9-9e46-128d03b4700e" "b013a134-06a6-31a9-9e46-128d03b4700e" "b013a134-06a6-31a9-9e46-128d03b4700e" ...
 $ course_number: chr  "224" "224" "224" "224" ...
 $ course_name : chr  "Introductory Statistics for Engineers" "Introductory Statistics for Engineers" "Introductory Statistics for Engineers" ...
 $ a_count     : chr  "17" "21" "15" "15" ...
 $ ab_count    : chr  "14" "16" "3" "11" ...
 $ b_count     : chr  "27" "23" "28" "27" ...
 $ bc_count    : chr  "7" "6" "6" "7" ...
 $ c_count     : chr  "20" "13" "11" "11" ...
 $ d_count     : chr  "2" "1" "1" "3" ...
 $ f_count     : chr  "0" "2" "2" "0" ...
```

```
dados_disc[cols_notas] <- lapply(dados_disc[cols_notas], as.numeric)
```

```
calcula_gpa <- function(df) {
  df$pontos <- 4 *df$a_count + 3.5*df$ab_count + 3*df$b_count +
              2.5*df$bc_count + 2 *df$c_count + 1*df$d_count +
              0 *df$f_count
  df$alunos <- df$a_count + df$ab_count + df$b_count + df$bc_count +
              df$c_count + df$d_count + df$f_count
  df
}

dados_prof <- calcula_gpa(dados_prof)
dados_disc <- calcula_gpa(dados_disc)

# Agregando por professor
por_prof <- aggregate(cbind(pontos, alunos) ~ instructor_id + instructor_name,
                      data = dados_prof, sum)
```

```
por_prof$gpa_medio <- por_prof$pontos / por_prof$alunos

# professor mais difícil (menor GPA médio) e mais fácil (maior)
por_prof[which.min(por_prof$gpa_medio), ]
```

|    | instructor_id | instructor_name | pontos | alunos | gpa_medio |
|----|---------------|-----------------|--------|--------|-----------|
| 72 | 2601633       | JAMES D KUELS   | 105.5  | 40     | 2.6375    |

```
por_prof[which.max(por_prof$gpa_medio), ]
```

|    | instructor_id | instructor_name | pontos | alunos | gpa_medio |
|----|---------------|-----------------|--------|--------|-----------|
| 64 | 4272783       | GUANHUA CHEN    | 64     | 16     | 4         |

```
# Agregando por disciplina
por_disc <- aggregate(cbind(pontos, alunos) ~ course_uuid + course_number + course_name,
                      data = dados_disc, sum)
por_disc$gpa_medio <- por_disc$pontos / por_disc$alunos

# disciplina mais difícil e mais fácil
por_disc[which.min(por_disc$gpa_medio), ]
```

|    | course_uuid                          | course_number | course_name                               | pontos | alunos | gpa_medio |
|----|--------------------------------------|---------------|-------------------------------------------|--------|--------|-----------|
| 28 | 3ef3ba60-426e-39ad-a61f-17d381b193d6 | 431           |                                           |        |        |           |
| 28 |                                      |               | Introduction to the Theory of Probability | 6439.5 | 2231   | 2.886374  |

```
por_disc[which.max(por_disc$gpa_medio), ]
```

|   | course_uuid                          | course_number | course_name            | pontos | alunos | gpa_medio |
|---|--------------------------------------|---------------|------------------------|--------|--------|-----------|
| 7 | 74fb97c9-9e98-3e8a-b0d9-66123b5a8e33 | 628           | Data Science Practicum |        |        |           |
| 7 |                                      |               |                        | 48     | 12     | 4         |

#### 04) Desconectar o banco

```
dbDisconnect(conn)
```

## Horario de compilação

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-23 22:53:59 -03"
```